Pamela Peterson Drake PhD CFA

Pamela Peterson Drake, PhD, CFA, is at James Madison University (USA).

SIMPLE LINEAR REGRESSION

LEARNING OUTCOMES

The candidate should be able to:

- describe a simple linear regression model, how the least squares criterion is used to estimate regression coefficients, and the interpretation of these coefficients
- explain the assumptions underlying the simple linear regression model, and describe how residuals and residual plots indicate if these assumptions may have been violated
- calculate and interpret measures of fit and formulate and evaluate tests of fit and of regression coefficients in a simple linear regression
- describe the use of analysis of variance (ANOVA) in regression analysis, interpret ANOVA results, and calculate and interpret the standard error of estimate in a simple linear regression
- calculate and interpret the predicted value for the dependent variable, and a prediction interval for it, given an estimated linear regression model and a value for the independent variable
- describe different functional forms of simple linear regressions

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